

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

*I Abhinav Charles 192524168 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Scenario 1: Smart Campus Subnet Allocation

- **Base Network Block:** 192.168.100.0/24
- **Allocation strategy:** To avoid overlapping, allocate addresses starting from the largest subnet block (/25) down to the smallest (/27).

Subnet Allocation Table

Department	Subnet Mask	Network Address	Usable Host IP Range	Broadcast Address	Total Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical	/27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

4. Remaining Unused IP Address Range

- The last allocated address is 192.168.100.223.
- The entire block ends at 192.168.100.255.
- **Unused Range:** 192.168.100.224 to 192.168.100.255 (Equivalent to a /27 and a /28 block, or broken into remaining smaller subnets).

Scenario 2: Multinational Software Company Subnet Allocation

- **Base Network Block:** 10.10.0.0/16
- **Allocation strategy:** Order the departments sequentially from largest allocation requirement (/24) to smallest (/27) to maintain optimal contiguous blocking.

Subnet Allocation Table

Department	Subnet Mask	Network Address	Usable Host IP Range	Broadcast Address	Total Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126

Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

4. Remaining Unused IP Address Range

- The allocated subnets completely occupy up to IP address 10.10.1.223.
- **Unused Range:** 10.10.1.224 through 10.10.255.255.

Question 3: IPv6 Address Configuration

Given Network Address: 2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

- Omit leading zeros in each 16-bit block.
- Replace consecutive blocks of zeros with a single double colon (: :). This substitution can only be performed once in an address.
- **Answer:** 2001:db8:abcd::/64

2. Full Hexadecimal Representation

- Expand the compressed parts back to their original 8 blocks of 4 hexadecimal digits (32 bits per block, 128 bits total).
- **Answer:** 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix

- The /64 prefix length specifies that the first 64 bits represent the network portion.
- Each block contains 16 bits, so the first 4 blocks ($16 * 4 = 64$) form the network prefix.
- **Answer:** 2001:0db8:abcd:0000::/64 (or 2001:db8:abcd::/64)

4. Total Number of Host Addresses Supported Within the Subnet

- An IPv6 address has 128 bits total.
- The host portion has $(128 - 64 = 64)$ bits.
- **Answer:** (2^{64}) (which equals 18,446,744,073,709,551,616) addresses

5. Total Number of Possible Host Addresses Available Within This Network

- Unlike IPv4, standard IPv6 subnets do not subtract addresses for network and broadcast boundaries. All (2^{64}) addresses are usable for host assignments.
- **Answer:** (2^{64}) addresses

Question 4: Routing Table Operations

Routing Table Entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Incoming Packet Destination IP: 192.168.2.45

1. The destination subnet to which the packet belongs.

- A /24 prefix length means the first 24 bits (the first 3 octets) represent the network id.
- The first 3 octets of the destination IP are 192.168.2.
- **Answer:** 192.168.2.0/24

2. The matching routing table entry.

- The destination network matches the second row in the provided table exactly.
- **Answer:** 192.168.2.0/24

3. The next-hop network/interface used to forward the packet.

- The router will use the outbound interface or next-hop IP bound specifically to the matching entry. Since specific interface labels (like Fa0/1 or Serial0) are not explicitly named in the table text, it points directly to the subnet path.
- **Answer:** The interface associated with the 192.168.2.0/24 network link.

4. The default route that should be used if the destination does not match any routing table entry.

- The standard default route notation matches all destinations not specified elsewhere.
- **Answer:** 0.0.0.0/0

5. Administration LAN (192.168.10.0/26) Network Address: 192.168.10.0 Broadcast Address: 192.168.10.63 Valid Host IP Address Range: 192.168.10.1 to 192.168.10.62 Suggested IP Addresses for Three Computers: 192.168.10.2, 192.168.10.3, and 192.168.10.4 (Note: 192.168.10.1 is already assigned to the gateway) Default Gateway: 192.168.10.1 Finance LAN (192.168.10.64/26) Network Address: 192.168.10.64 Broadcast Address: 192.168.10.127 Valid Host IP Address Range: 192.168.10.65 to 192.168.10.126 Suggested IP Addresses for Three Computers: 192.168.10.66, 192.168.10.67, and 192.168.10.68 (Note: 192.168.10.65 is already assigned to the gateway) Default Gateway: 192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand